

	
8(a)	From Fig. 8.1, the period of the pulsar = 0.60 s Angular velocity, $\omega = \frac{2\pi}{T}$ $= \frac{2\pi}{0.60}$ $= 10.472$ $\approx 10.5 \text{ rad s}^{-1}$
8(b)	Using $L = \frac{2}{5}mr^2\omega = \frac{2}{5}mr^2\left(\frac{2\pi}{T}\right)$, where T is the period of rotation. Since angular momentum L is conserved in a closed system, the period of rotation T is directly proportional to the square of radius r of the pulsar. As the pulsar has a smaller radius than that of the star from which it was formed, the period of rotation T is smaller, when it spins much faster. [This question proves challenging for many candidates.]
8(c)	The <u>nucleus of iron</u> cannot undergo nuclear fusion to release energy because it has the highest binding energy per nucleon of any element. This means that the nucleus of iron is very stable and it will actually require more energy to fuse iron nuclei together than the energy that would be released in the process.

	[This question proves challenging for many candidates in terms of using the correct terminology of the nucleus of iron instead of the element or atom of iron.]
8(d)(i)	Using $g = \frac{GM}{r^2}$, $g = \frac{(6.67 \times 10^{-11})(1.4 \times 2.0 \times 10^{30})}{(10 \times 10^3)^2}$ $= 1.868 \times 10^{12}$ $\approx 1.87 \times 10^{12} \text{ N kg}^{-1}$
8(d)(ii)	$\Delta g = g_{\text{surface}} - g_{1\text{m above surface}}$ $= (6.67 \times 10^{-11})(1.4 \times 2.0 \times 10^{30}) \left[\frac{1}{(10 \times 10^3)^2} - \frac{1}{(10 \times 10^3 + 1)^2} \right]$ $= 3.735 \times 10^8$ $\approx 3.74 \times 10^8 \text{ N kg}^{-1}$
8(d)(iii)	The length of the rod will lengthen as it accelerates towards the surface of the pulsar because the difference in gravitational force over one metre pulling on the top and bottom of the rod is very large, with the bottom end of the rod experiencing the larger gravitational force. [This question proves challenging in the demand of clarify of either the gravitational field strength or gravitational force on the top and bottom end of the rod.]
8(e)(i)	The speed of the surface of the pulsar cannot exceed the speed of light, which the linear velocity ($v = r\omega$) of the surface of the pulsar, where r is the radius of the pulsar and ω is the angular speed of the pulsar. i.e. $r\omega < c$ $r \left(\frac{2\pi}{T} \right) < c$ $T > \frac{2\pi r}{c}$ $= \frac{2\pi \times 100 \times 10^3}{3.00 \times 10^8}$ $= 0.00209$ $\approx 2.1 \text{ ms}$ Hence, the pulsar cannot have a period of 2.0 ms.
8(e)(ii)	For the spherical pulsar not to disintegrate, the gravitational field strength on the surface of the pulsar must be equal to the centripetal acceleration at the surface. i.e. $g = a_c$

	$\frac{GM}{R^2} = R\omega^2$ where M is mass of pulsar R is radius of pulsar ω is angular speed of rotation $GM = R^3 \left(\frac{2\pi}{T} \right)^2$ $M = \frac{4\pi^2 R^3}{GT^2}$ Since, density $\rho = \frac{M}{V}$, where V is volume of sphere $\rho = \frac{\frac{4\pi^2 R^3}{GT^2}}{\frac{4}{3}\pi R^3}$ $= \frac{3\pi}{GT^2} \text{ (shown)}$
8(e)(iii)	Using $\rho = \frac{3\pi}{GT^2}$, $\rho = \frac{3\pi}{(6.67 \times 10^{-11})(0.60)^2}$ $= 3.925 \times 10^{11}$ $\approx 3.93 \times 10^{11} \text{ kg m}^{-3}$
8(f)	Using age = $\frac{\text{period}}{2 \times \text{rate of change of period}}$, Age of the Crab Pulsar $= \frac{33.5 \times 10^{-3}}{2 \times 38 \times 10^{-9}}$ $= 440789 \text{ days}$ $= 1208 \text{ years}$ $\approx 1210 \text{ years}$ [Many candidates did not consider the units of their answer, and left their answers in days rather than years. Many candidates did not use the given expression and the data given in the text.]
8(g)	Using $E = \frac{hc}{\lambda}$,

	$\lambda = \frac{hc}{E_{\max}}$ $= \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{(400 \times 10^{12})(1.60 \times 10^{-19})}$ $= 3.108 \times 10^{-21}$ $\approx 3.11 \times 10^{-21} \text{ m}$ [Many candidates were unable to convert the energy of the photons in TeV into joules.]
8(h)	The intensity of the radio wave signal of a pulsar arriving on Earth is very weak. Since the power of the received signal is directly proportional to the collecting area of the detector. The effective collecting area of SKA of 1.0 km^2, i.e. $1,000,000 \text{ m}^2$, is more than 300 times larger than that of the Parkes Telescope ($\pi \times 32^2 = 3200 \text{ m}^2$). [Many candidates interpreted the larger collecting area to mean being able to "see" more of the sky, rather than relating to the concept of intensity and power of signal received.]